

Activity 14 Instruction Manual

Activity 14 | Topic 02 | C013 | Version v1.0

Purpose

Having found a root cause, you need a disciplined way to fix it and make the fix stick. You will run the same improvement through both PDCA and DMAIC and see where they differ.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A completed PDCA cycle and a DMAIC project definition for your improvement.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. Take the root cause you identified in Activity 13. Open 'Plan Do Check Act.pptx'.
2. Prompt GenAI: 'Design a PDCA cycle to address [root cause]. For Plan, state the data reviewed and the prioritised solutions; for Do, define a small pilot; for Check, define the measure of success; for Act, state what happens if it works and if it does not.'
3. Sharpen the Check stage - this is where AI drafts are weakest. Insist on a specific number and a date: 'rework hours per sprint below 8 by 30 September', not 'improved efficiency'.
4. Complete the PDCA template. Confirm your Do stage is genuinely a small pilot rather than a full rollout.
5. Now open 'DMAIC.pptx' and prompt: 'Structure the same improvement as a DMAIC project. Define the gap, state how I would measure the current problem with real data, analyse root cause, propose the improvement, and specify the control that keeps the process in place.'
6. Compare the two: PDCA is a fast learning loop; DMAIC is a heavier, data-driven method for a chronic problem. State which fits your improvement and why.
7. Pay attention to the Control stage - ask GenAI 'what would cause this improvement to decay after six months?' and build a control against it.
8. Share your improvement plan with the class.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- Your Check/Control stages specify a number and a date, your Do stage is a pilot, and you can justify choosing PDCA or DMAIC for this problem.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.